

AKSHAR VANDARA

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EDUCATION

Northeastern University, Boston, MA

May 2026

Master of Science, Game Science and Design, GPA 3.87/4

Relevant Courses: Game Data Science, Mixed Research Methods in Games, Generative Game Design

Gujarat Technological University, Ahmedabad, India

Jun 2023

Bachelor of Engineering, Computer Engineering, GPA 3.6/4

Relevant Courses: Operating System, Software Engineering, Artificial Intelligence, Cloud Computing

TECHNICAL SKILLS

- **Game Dev:** Unity, Unreal Engine, Blueprints, Game Physics, Core-gameplay management, Database handling
- **Game Design:** Animations, Animation Rigging, UI integration, Environment Design
- **Programming Languages:** C#, C++, JavaScript, Python, Java, R, C, SQL
- **Python Libraries:** NumPy, Pandas, PyTorch, Cuda, TensorFlow, PyGames, Flask, Django, SQLAlchemy
- **Front-End:** HTML, CSS, JavaScript, Three.js, Tailwind, React, Node, Vue, Angular, Next.js, Vite.js
- **Developer Tools:** Git, Perforce, Firebase Realtime DB, Firebase Cloud functions, SourceTree, Gitlab, Figma, Trello

PROFESSIONAL EXPERIENCE

Research Assistant

Northeastern University, Boston, MA, USA

Feb 2025 – Present

- Extended the 'Sturgeon' constraint-solving system to generate 2D puzzle levels along with their playthroughs.
- Improved the generation algorithm's performance by over 100x and increased its reliability over prior academic methods.
- Co-authored a paper detailing the system's architecture and results, published at the AAAI 2025 AIIDE conference.
- Currently working on generating 3D game levels by applying patterns extracted from 2D games to the system.

Unreal Developer

Otisco Studios, FL, USA

May 2025 – Aug 2025

- Engineered core mechanics for Cricket Tycoon, using state machines and player interaction systems, achieving a 30% improvement in gameplay responsiveness.
- Developed 8+ core gameplay systems for two titles, covering save management, UI, and environment mechanics.
- Implemented AI for enemies and NPCs, delivering 5+ archetypes in roguelike and open-world genres using Behavior Trees and Blackboards.
- Identified and fixed over 30+ bugs prior to launch, directly improving product stability.
- Profiled and refactored code across multiple projects, increasing average FPS by ~100% during gameplay.

Unity Developer

Arcadon Games, Bangalore, India

Mar 2023 – Jul 2024

- Engineered core mechanics for Cricket Tycoon, using state machines and player interaction systems, achieving a 30% improvement in gameplay responsiveness.
- Integrated Firebase backend solutions, managing over 1,000+ user data entries, inventory systems, and game progress synchronization with real-time updates via cloud functions.
- Designed AI simulation systems capable of simulating up to 30 randomized matches per minute, enhancing gameplay variety and reducing load times by 40%.
- Conducted alpha testing with 100+ users, achieving 85% positive feedback, and resolved 20+ critical issues to significantly improve game stability and user engagement.

PROJECTS

Spacetime Level Generation [Link](#)

Feb 2025 – Present

- Modified a Python constraint-solving system ('Sturgeon') to generate 2D puzzle game levels along with their solutions.
- Increased level generation speed by 100-500 times and improved error-proofing compared to previous methods like STWFC (Space-Time WaveFunctionCollapse)
- Co-authored and published a research paper on the system that was accepted into the 2025 AAAI conference (AIIDE).

Red Circuit Rebellion [Link](#)

Dec 2024 – April 2025

- Developed 5+ core gameplay systems, including save/load, environmental fracturing, health management, and objectives manager.
- Implemented enemy AI using Behavior Trees and EQS to create 3 combat archetypes.
- Built 2 game modes (Story and Arcade) with specific objective logic and enemy spawners for each.
- Optimized game assets and blueprints, increasing average FPS by ~100% during gameplay.

Cricket Manager: League [Link](#)

Mar 2023 – Jul 2024

- Built sports management system capable of handling 1000+ concurrent users with Firebase, maintaining sub-100ms response times for pseudo-multiplayer features.
- Orchestrated alpha testing phase with 100+ users, iterating through 3 major versions to achieve 85% user satisfaction and 40% reduction in bug reports.